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app = workflow.compile()
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Processing user queries

A list of user queries is defined. These are processed one by one as follows.

- The workflow is invoked with the query as input.
- The final response is obtained and processed.
- Time taken for each query is logged.

```
# List of user queries (Very definite questions)
user_queries = [
    "What's the weather like today?",
    "Tell me the latest news.",
    "Can you tell me a joke?",
    "What is today breaking news in Kolkata?"
]

for idx, query in enumerate(user_queries, start=1):
    print(f"\n--- Processing Query {idx}/{len(user_queries)} ---")

    start_time = time.time()
    final_state = app.invoke(ExpertSystemState(input_data=query))
    final_state = ExpertSystemState(**final_state)
    end_time = time.time()

    log_event("Response Time", f"{round(end_time - start_time, 2)} seconds")
    print("\nFinal Response:")
    print(f"- User Query: {final_state.input_data}")
    print(f"- Category: {final_state.category}")
    print(f"- Final Response: {final_state.response}")
    print("-" * 50)
```

The output is:

```
--- Processing Query 1/4 ---
[Super Agent] Analyzing query: What's the weather like today?
[Super Agent] Classified as: weather_service
[Weather Service Agent] Providing weather forecast for:
What's the weather like today?
[Weather Service Agent] Generated Response: The weather
forecast for today is sunny with a high of 25°C and a low of
15°C.
[Response Time] 0.42 seconds

Final Response:
- User Query: What's the weather like today?
- Category: weather_service
- Final Response: The weather forecast for today is sunny
with a high of 25°C and a low of 15°C.
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--- Processing Query 2/4 ---
[Super Agent] Analyzing query: Tell me the latest news.
[Super Agent] Classified as: news_service
[News Service Agent] Providing latest news for: Tell me the
latest news.
[News Service Agent] Generated Response: The latest news is
that the stock market has reached an all-time high today.
[Response Time] 0.48 seconds

Final Response:
- User Query: Tell me the latest news.
- Category: news_service
- Final Response: The latest news is that the stock market has
reached an all-time high today.
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--- Processing Query 3/4 ---
[Super Agent] Analyzing query: Can you tell me a joke?
[Super Agent] Classified as: joke_service
[Joke Service Agent] Telling a joke for: Can you tell me a
joke?
[Joke Service Agent] Generated Response: Why don't scientists
trust atoms? Because they make up everything!
[Response Time] 0.43 seconds

Final Response:
- User Query: Can you tell me a joke?
- Category: joke_service
- Final Response: Why don't scientists trust atoms? Because
they make up everything!
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--- Processing Query 4/4 ---
[Super Agent] Analyzing query: What is today breaking news in
Kolkata?
[Super Agent] Classified as: news_service
[News Service Agent] Providing latest news for: What is today
breaking news in Kolkata?
[News Service Agent] Generated Response: Breaking news in
Kolkata: A new metro line has been inaugurated today, enhancing
the city's public transport system.
[Response Time] 0.45 seconds

Final Response:
- User Query: What is today breaking news in Kolkata?
- Category: news_service
- Final Response: Breaking news in Kolkata: A new metro line
has been inaugurated today, enhancing the city's public
transport system.
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▪ The Super Agent categorises each query (weather,
news, or joke).
▪ Based on the classification, the query is routed to the
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